AE 158 – Introduction to Information Technology (2,1,2)

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KNUST

LECTURE TWO

COMPONENTS OF AN INFORMATION SYSTEM

Objectives

- *To know the parts of an information system*
- *To understand the terms*
  - *hardware*
  - *Software*
- *To distinguish between systems software and applications software*
- *To give examples of several major types of application and system software*
- *To outline the functions of an operating system*

INFORMATION SYSTEM

An information system is integrated and co-ordinate network of components, which combine together to convert data into information

Data is facts and statistics collected together for reference or analysis.

Information is processed data into a context.

PARTS OF AN INFORMATION SYSTEM

- FIVE PARTS
  - **PEOPLE**
  - **SOFTWARE**
  - **HARDWARE**
  - **DATABASE**
  - **NETWORKS**

PEOPLE

- Two types of people use information technology

1. Computer professionals or information technology professionals (network administrator, system specialist)

2. End users

SOFTWARE

- **describe all the various programs that may be used on a computer system together with the associated documentation**
- the **programs** control the computer and make it function
  - Program is a set of instructions that the computer obeys
- Software is divided into two broad categories:
  - **systems software** and
  - **application software**.

- Manuals and guidelines
  - Instruct end users on how to use the software and hardware
  - They are descriptions of how things are done-steps for accomplishing a result or rules and guidelines for what is acceptable

Systems software

- Starts up the computer and functions as the principal coordinator of all hardware components.

- Without System software loaded into the computer's RAM, hardware and applications software are useless.

Systems software

- used to describe programs that enable the computer to function, improve its performance and access the functionality of the hardware
  - tells the computer how to communicate with peripheral devices
- It also provides support to the physical components of computers
- Sole function is the control of the operation of the computer
- Two major functional categories of system software include:
  - System management programs
  - System development programs

System management programs

- programs that manage the hardware, software, and data resources of the computer system during its execution of the various information processing jobs of users

- They include:
  - Operating Systems
  - Network Management Programs
  - Database Management Systems
  - System Utilities

Operating systems

- operating system actually makes the computer function
- is an integrated system of programs that:
  - manages the operations of the CPU
  - controls the input/output and storage resources and activities of the computer system
  - provides various support services as the computer executes the application programs of users.

Operating system functions

- performs five basic functions in the operation of a computer system. It provides:
  - User Interface
  - Resource Management
  - Task Management
  - File Management
  - Utilities and Support Services

Popular Operating Systems

- most popular microcomputer operating systems are
  - **MS-DOS**
  - **Windows 95**
  - **Windows 98**
  - **Windows 2000**
  - **Windows NT**
  - **UNIX**
  - **LINUX**
  - **Mac OS X**

Network management programs

- information systems rely heavily on the Internet, local area networks, and other telecommunications networks to interconnect
- system software for network management

Database Management Systems

- set of software programs that control the
  - development,
  - use, and
  - maintenance of the databases of computer-using organizations

Other system management programs

- performs routine support functions for the end users of a computer system. Examples are
  - **Utility programs**
    - perform miscellaneous housekeeping and file conversion functions
  - **Performance Monitors**
    - monitor the performance and usage of computer systems to help their efficient use
  - **Security Monitors**
    - monitor and control the use of computer systems and provide warning messages and record evidence of unauthorized use of computer resources

Applications software

- programs enable users to achieve specific objectives such as
  - create a document,
  - create a database,
  - produce a spreadsheet or
  - design a building
- programs designed to satisfy a user's needs
- consists of a variety of programs subdivided into
  - general-purpose and
  - application-specific categories

Application specific programs

- direct the processing required for a particular use, or application, that end users want accomplished

- Examples include:
  - Business Applications
  - Science and Engineering Applications
  - Education and Entertainment Applications
  - Personal Applications

General-purpose application programs

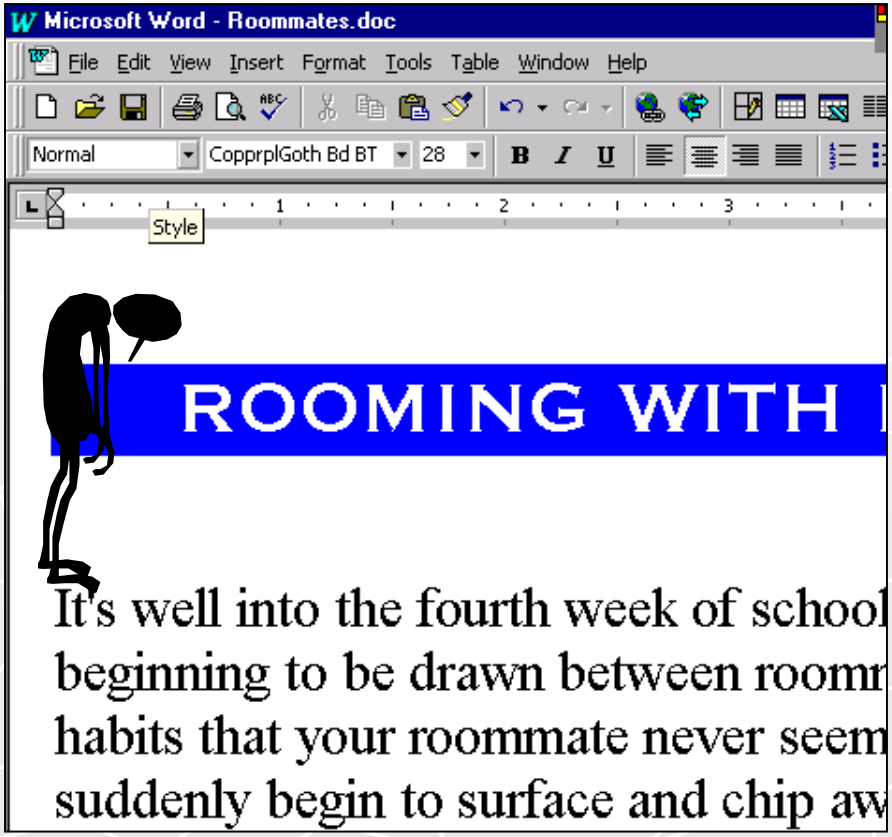
- perform common information processing jobs for end users.
- Examples include:
  - Word processing programs
  - Spreadsheet programs
  - Database management programs
  - Graphics programs
- categorised into
  - basic tools and
  - power tools

Basic Tools	Power Tools
Word Processors	Personal Information Managers
Spreadsheet Programs	Groupware
Database Managers	Project Managers
Graphics Programs	Multimedia
Integrated Programs	Artificial Intelligence
Desktop Publishing	

Basic Tools

- **Word Processing**
  - Deals with creation, editing, revision, and printing of documents by electronically processing text data
  - features of these packages include:
    - Desktop publishing capabilities
    - spell checker, thesaurus, grammar and punctuation capabilities
    - Suggest possible improvements in your writing style with grammar and style checker functions
    - Design and create pages from scratch for an Internet or intranet web site

- Creating & saving a document
- Formatting
  - Typefaces
  - Fonts
- Clip art



Desktop Publishing (DTP)

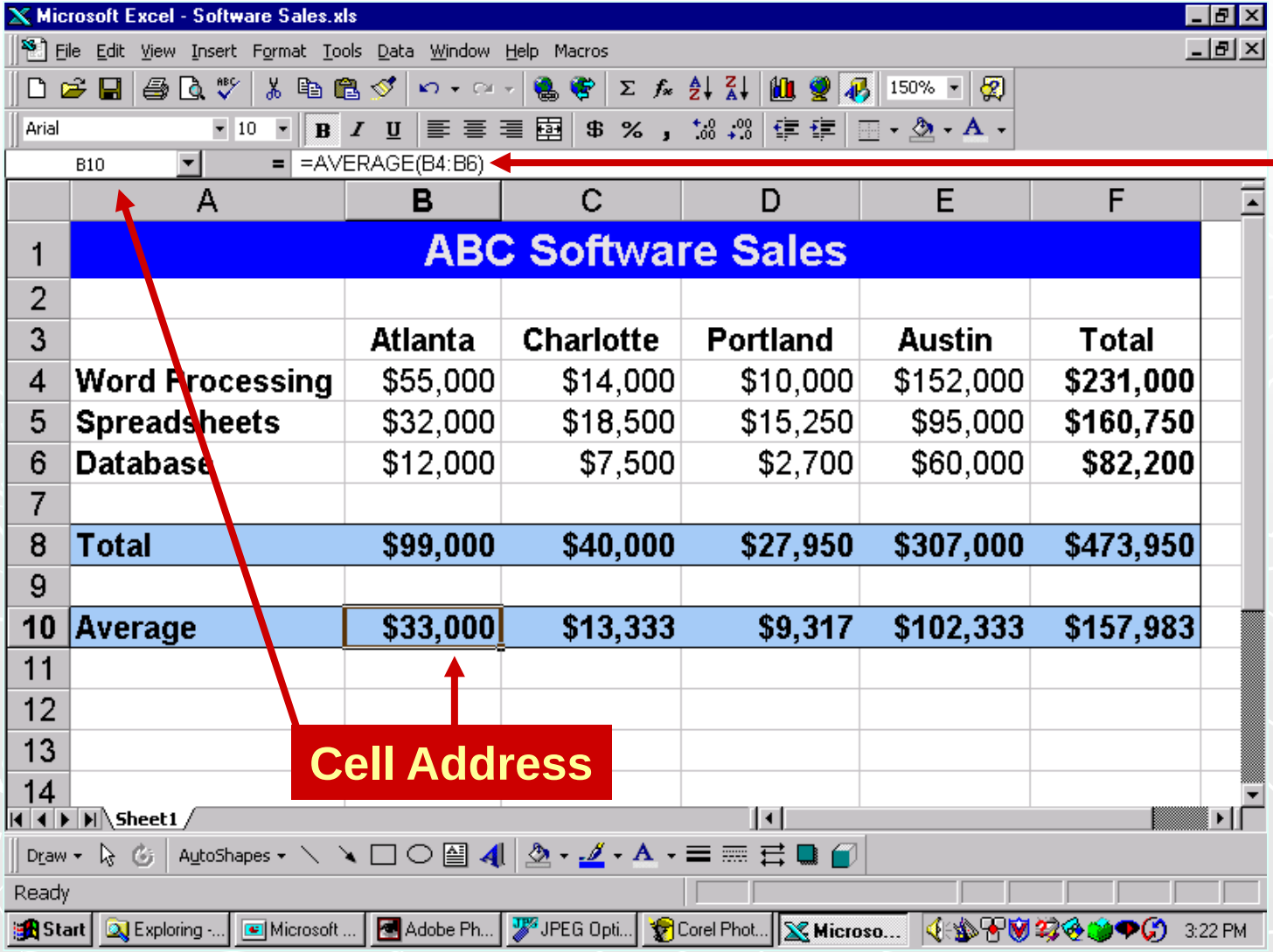
- Enables end users produce their own
  - printed materials such as newsletters, brochures, manuals, and books with several type styles, graphics, and colours on each page

- heart of desktop publishing is a page design process called *page markup* or *page composition*.

Electronic Spreadsheets

- used for business analysis, planning, and modelling
- Principal features
  - Format: Labels (column and row headings) and values (numbers in cells) can be displayed in different ways.
  - Formulas and Functions: Formulas are instructions for calculations
  - Functions built-in formulas.
    - Recalculation: Recalculation means automatic recomputation
    - Analysis Tools: Built-in tools for what-if, goal-seeking, solving, and scenario analysis.

Spreadsheet Software



Cell Address

Cell Range and Cell Entries

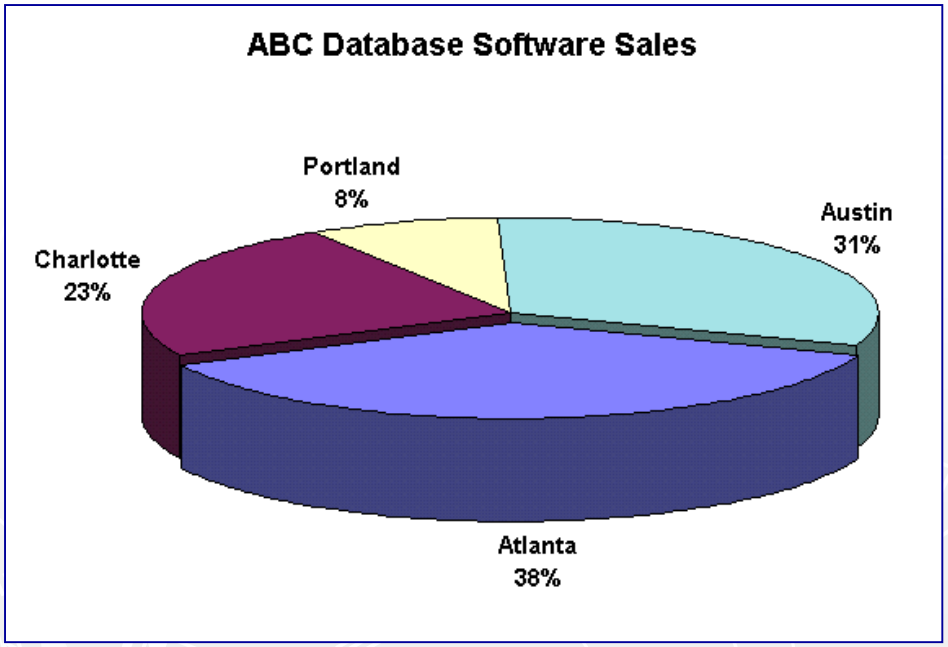
Rows and Columns

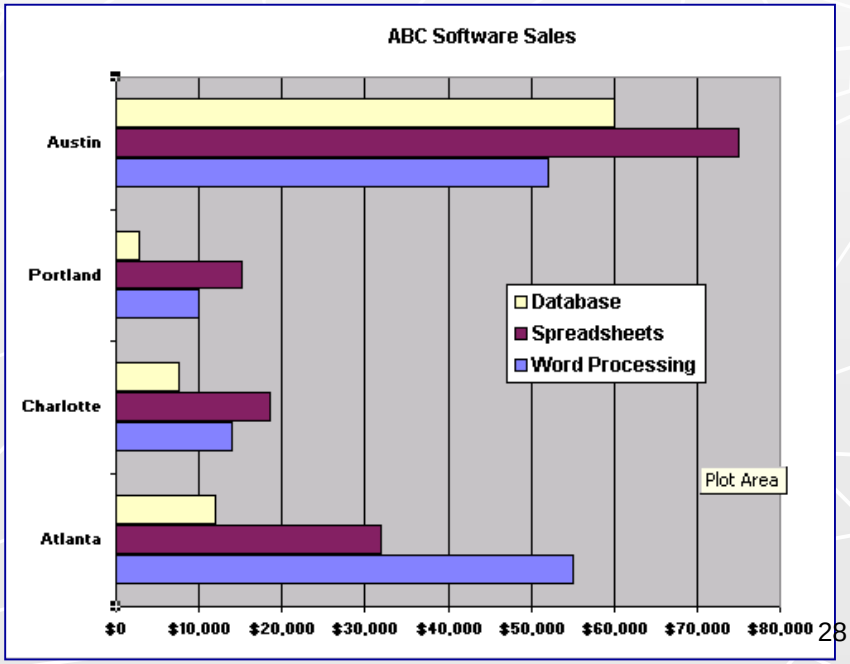
Functions

Templates

Spreadsheet/ Database

Spreadsheet Graphics





Presentation Graphics

- used to convert numeric data into graphics displays such as line charts, bar graphs, pie charts, and many other types of graphic

- Three types of graphics programs
  - analytical,
  - presentation and
  - drawing programs.

Analytical Graphics

- programs put data in a form easier to analyse.
- Example:
  - as bar charts and line graphs.

Presentation Graphics

- communicate results or persuade, using colour and titles.

Drawing Programs

- professional illustrators create attractive, sophisticated artwork for publications.

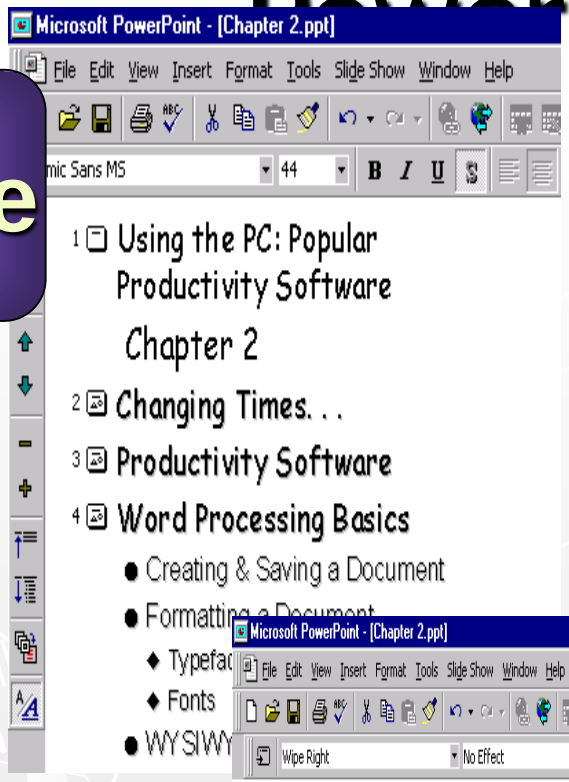
Example Packages

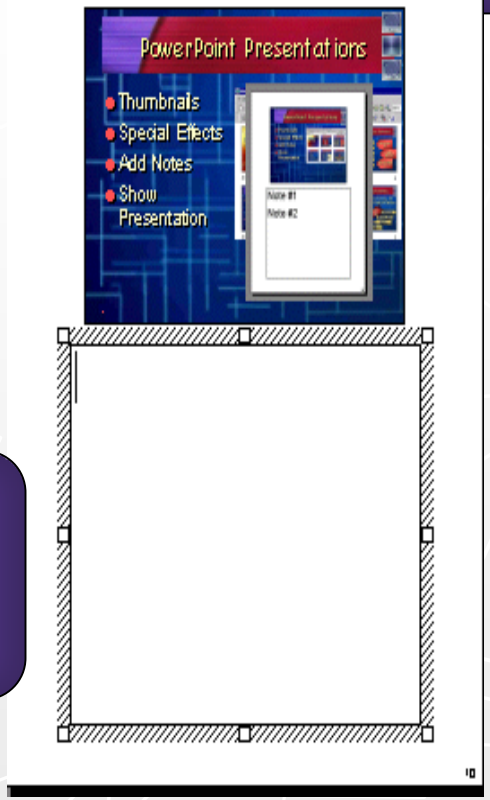
- Most analytical graphics come as part of spreadsheet programs.
- Presentation Graphics
  - Lotus Freelance Graphics
  - Harvard Graphics
  - Persuasion
  - PowerPoint
- Drawing Programs
  - Adobe Illustrator
  - Aldus Freehand
  - Micrografix Designer

PowerPoint Presenta

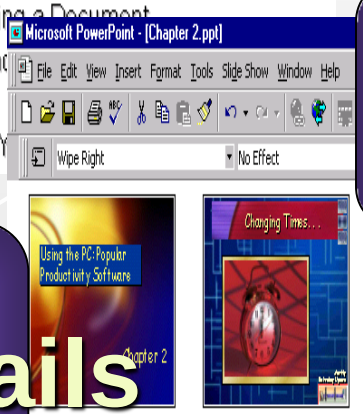
Special Effects

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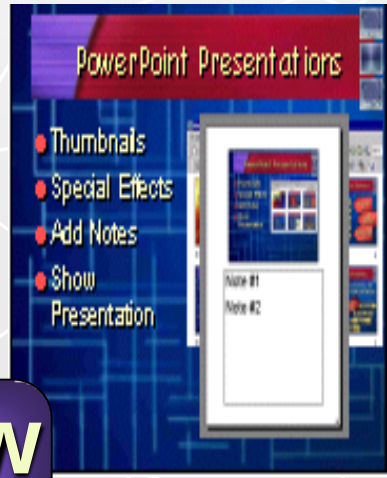


Notes Pages



umbnails





Show Presentation

DATABASE MANAGERS or DATABASE MANAGEMENT SYSTEM, DBMS

- used to structure a database.
  - A database is a collection of related data
  - Relational databases are organised as fields, records, and tables for easy retrieval of selected items.
- Most DBMS packages can perform four primary tasks
  - Application Development
  - Database Interrogation
  - Database Maintenance
  - Database Development

Database Software

- Create and maintain a database
- Make inquiries
- Sort records
- Print reports with subtotals & totals



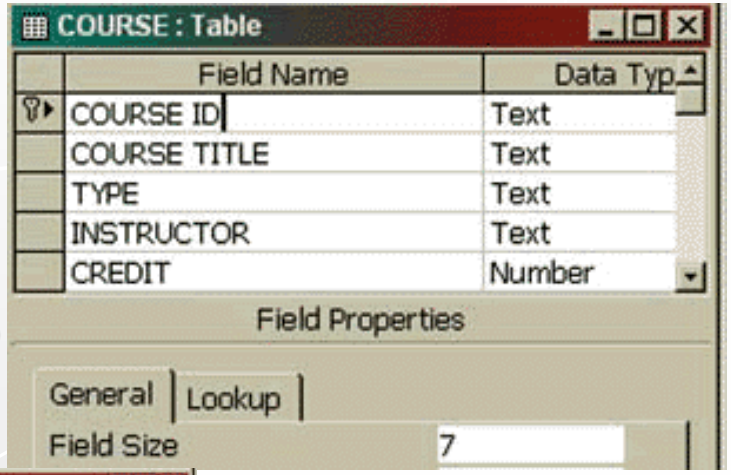
Database Software



Databases
vs.
Spreadsheets

Database Software

- Database
  - Fields
  - Records
  - Tables (files)
  - Relational Databases
- Database Structure



COURSE : Table

COURSE ID	COURSE TITLE	TYPE	INSTRUCTOR	CREDIT
CIS 100	Telecommunications	lecture	Babbage	3
CIS 101	Database Management	lecture/lab	Babbage	2
CIS 110	Intranets and Extranets	lab	Gates	3
CIS 150	Mgt. Info. Systems	lecture	Eckert	3
CIS 202	Visual Basic Programming	lecture/lab	Wang	4
CIS 310	Network Administration	lecture	Gates	4
CIS 320	Desktop Publishing	lab	Eckert	3
CIS 330	Information Retrieval	lecture/lab	Wang	3
CIS 350	Web Site Design	lab	Gates	1
CIS 401	Local Area Networks	lecture/lab	Wang	3

Record: 1 of 10

Software Suites and Integrated Packages

- Software suites
  - a combination of the most widely used productivity packages
  - They include suites such as Microsoft Office, Lotus SmartSuite, and Corel WordPerfect Office, and Sun's StarOffice
  - Some Features
    - Windows allows sharing of data between completely different applications programs.
    - Less expensive than buying each individual software package

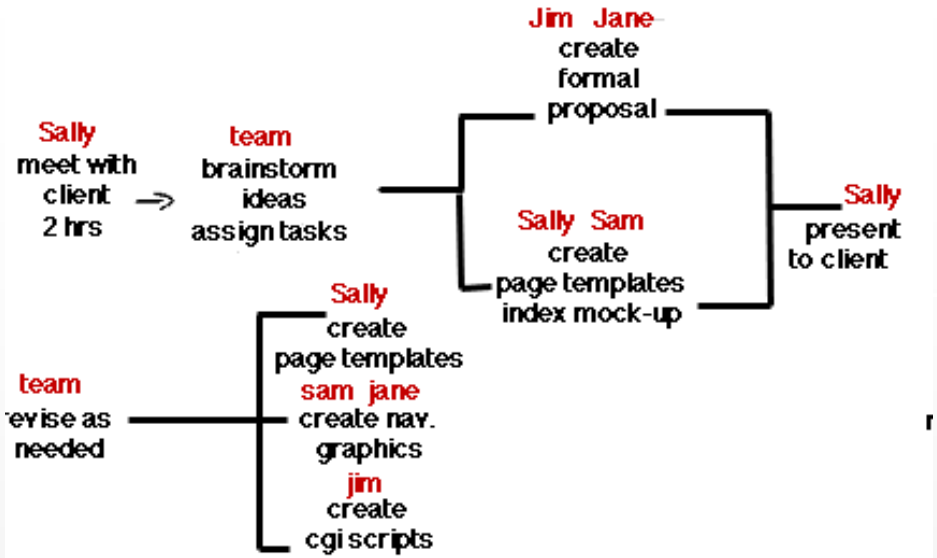
- Integrated packages
  - is an all-in-one software program.
  - Some Features
    - Common structure for easy data exchange between applications.
    - Less expensive than software suite.
    - Each application not as powerful as "stand alone" application

Power Tools

- **Personal Information Managers (PIMs)**
  - software package for end user productivity and collaboration
  - used to help end users
    - store, organize, and retrieve information about customers, clients, and prospects, or schedule and management appointments, meetings, and tasks.
- **Groupware**
  - is collaboration software that helps workgroups and teams work together to accomplish group assignments.

- **Project management**
  - allows one to plan, schedule, and control the people, resources, and costs of a project.
  - Two important tools found in project management software are:
    - Gantt Charts
    - PERT Charts
- **Multimedia**
  - integrates all sorts of media into an interactive presentation.
- **Artificial Intelligence**
  - is a research field to develop computer systems simulating human thought processes and actions.

Example of Gantt Chart and PERT Chart



Activity	Feb									Mar				
	21	22	23	24	25	26	27	28	29	1	2	3	4	5
meet with client	■	■	■	■	■									
develop Storyboard						■	■	■						
brainstorm ideas		■	■	■	■									
final proposal				■	■	■	■	■	■					
assign team jobs	■	■	■	■	■									
develop templates								■	■	■	■	■	■	■
create timeline									■	■	■	■	■	■
develop index page								■	■	■	■	■	■	■
photograph animals											■	■		
scan photos													■	■